Prof. Sunwoo Lee
Last Modified: 20260227

HOUSEKEEPING

- *Dr. Yamin Yan will take over from Week 8 and onwards (Tutorial & Assignment Problems are still me)*

EE2101 Circuit Analysis

Lecture 7

School of Electrical and Electronic Engineering
College of Engineering
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RECAP

- Capacitors and Inductors → Differential Equations → First and Second Order Circuits
- A frequency (Laplace) domain math can be easier than the time domain math:
  - Laplace Transform: Time Domain → Frequency Domain → “Re-arrangement / organization”
  - Inverse-Laplace Transform: Frequency Domain → Time Domain
- Changing circuits from Time → Laplace (C & L’s derivative nature leads to initial condition sources + source-free C & L)
- Transfer function denotes the ratio of output/input (instead of giving you the input value/number and asking for an output).

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Zeros, poles, and stability

- **Zeros, poles, and stability**
- **Complex number and its relationship to phase angle**

Zeros, poles, and stability

Laplace Transform involves "***s***"

s can be thought of as a generalized "frequency": s = ***σ* + j*ω***

s covers both the periodic sine waves (***ω***) and non-periodic exponentials (***σ***)

$$e^{st} = e^{(\sigma + j\omega)t} = \boxed{e^{\sigma t}}\boxed{e^{j\omega t}}$$

exponential

oscillatory

$$e^{-at} = \frac{1}{s+a} \; in\ Laplace\ (Frequency) Domain\ \&\ pole = -a$$

If –a < 0, stable:
"poles are on the left half of the plane (LHP)"

If –a > 0, unstable:
"poles are on the right half of the plane (RHP)"

Zeros, poles, and stability

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$$e^{st} = e^{(\sigma + j\omega)t} = \boxed{e^{\sigma t}}\boxed{e^{j\omega t}}$$

exponential

oscillatory

Only expoentials changes the "state"
(Recall how we saw system go from one value to another "exponentially")

In a steady state: σ = 0

Stability: The denominator of *H(s)* plays a critical role in "stability"

Ex) Tutorial 8 Problem 2, with zero initial conditions ($V_C(0) = 0$ & $I_L(0) = 0$)

$$\frac{i_o(s)}{i_g(s)} = \frac{s^2 + (2-2\mu)s}{\boxed{s^2 + (6-2\mu)s + 5}} \longrightarrow s^2 + (6-2\mu)s + 5 = 0 \text{ (Characteristic Equation)}$$

$$s_{1,2} = \frac{-(6-2\mu) \pm \sqrt{(6-2\mu)^2 - 20}}{2}$$

$$For\ stability, \frac{6-2\mu}{2} > 0\ , hence\ \ 0 < \mu < 3$$

WHY? $\boxed{L\{f(t) = e^{at}\} = \frac{1}{s-a}}$ *The exponential explodes if a > 0 & t > 0

*H(s)*input(s) = output(s) & Convolution in T-domain → Multiplication in F-domain*

$v_o(t) = h(t) \circledast v_g(t)$ Convolution in time domain (*i.e.*, a function is "applied" at the input)

$\Updownarrow$

$V_O(s) = H(s)V_g(s)$ Multiplication in frequency domain

Ex) in Tutorial 8 Problem 3, we used:

$$V_O(s) = H(s)V_g(s) = \left(-\frac{5000}{s+500}\right)\left(\frac{1}{s}\right)$$

By definition:
$H(s) = V_O(s)/V_g(s)$

$$= -10\left[\frac{1}{s} - \frac{1}{s+500}\right]$$

$$v_O(t) = \boxed{10[e^{-500t} - 1]\,u(t)}$$

You can get $v_O(t)$ by either

1) $v_o(t) = h(t) \circledast v_g(t)$

2) $V_O(s) = H(s)V_g(s)$ *then* $v_O(t) = L^{-1}\{V_O(s)\}$ → This is easier

7. s = jω (s = α + jω but for us α = 0 "steady state")

Ex) in Tutorial 8 Problem 4,

$$H(s) = \frac{V_O(s)}{V_g(s)} = \frac{s^2}{s^2 + 2000\,s + 4 \cdot 10^6}$$

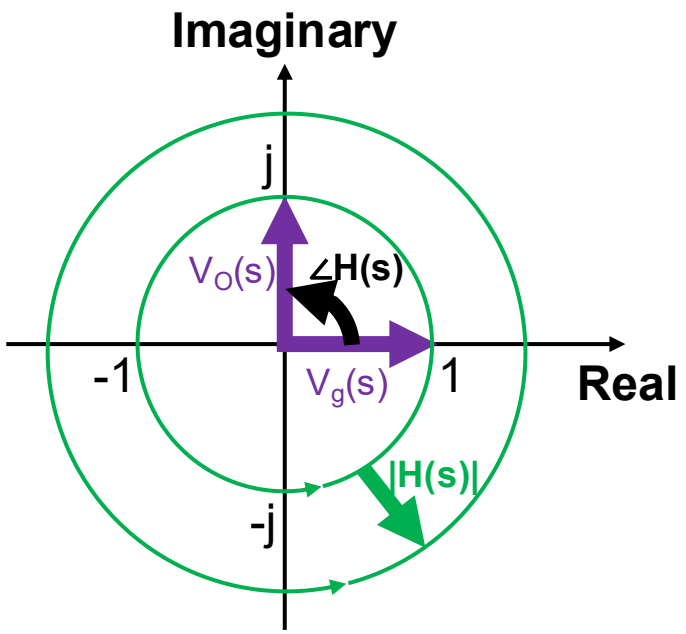
→ *This function describes the system behaviour (H(s)) for* ***all s***

**for all frequencies (s = jω = 2πf)*

if $V_g(t)$ = 10 cos (2000 t) means that ω = 2000

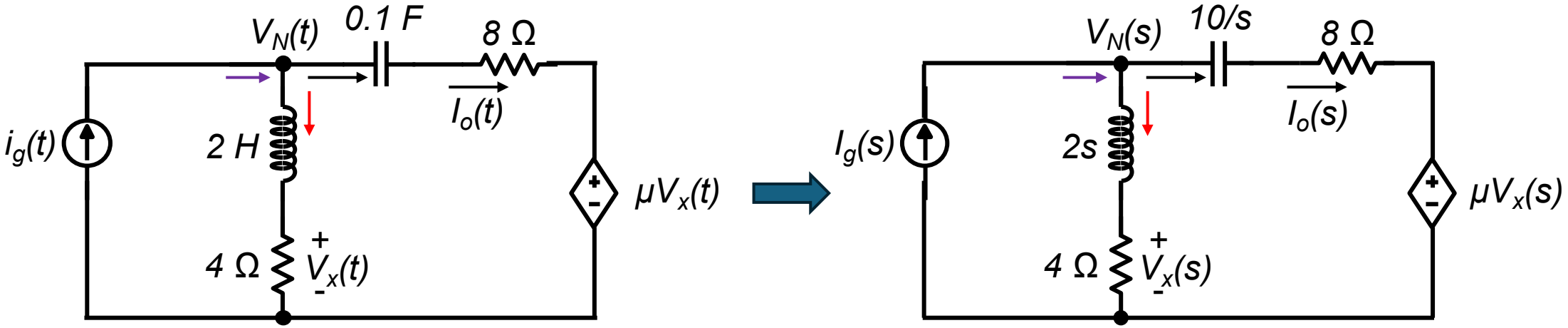
$$H(s = j2000) = \frac{(j2000)^2}{(j2000)^2 + 2000(j2000) + 4 \cdot 10^6} = j = 1\angle 90°$$

$$V_O(t) = |H(j2000)| \cos\big(\omega t + \angle H(j2000)\big) = 10\cos(\omega t + 90°)$$



Concepts: Laplace Transform with Zero Initial Conditions + Stability

With zero initial conditions (**$V_C(0) = 0$ & $I_L(0) = 0$**), find $H(s) = I_o(s)/I_g(s)$ → Laplace Transform



$$1 - \frac{i_o(s)}{i_g(s)} = \frac{(10+8s)}{2s^2+12s-4\mu s+10} = \frac{(5+4s)}{s^2+(6-2\mu)s+5}$$

$$\frac{i_o(s)}{i_g(s)} = 1 - \frac{(5+4s)}{s^2+(6-2\mu)s+5} = \boxed{\frac{s^2+(2-2\mu)s}{s^2+(6-2\mu)s+5}}$$

$$s^2+(6-2\mu)s+5=0 \quad \text{(Characteristic Equation)}$$

$$s_{1,2} = \frac{-(6-2\mu) \pm \sqrt{(6-2\mu)^2-20}}{2}$$

Concepts: Laplace Transform with Zero Initial Conditions + Stability

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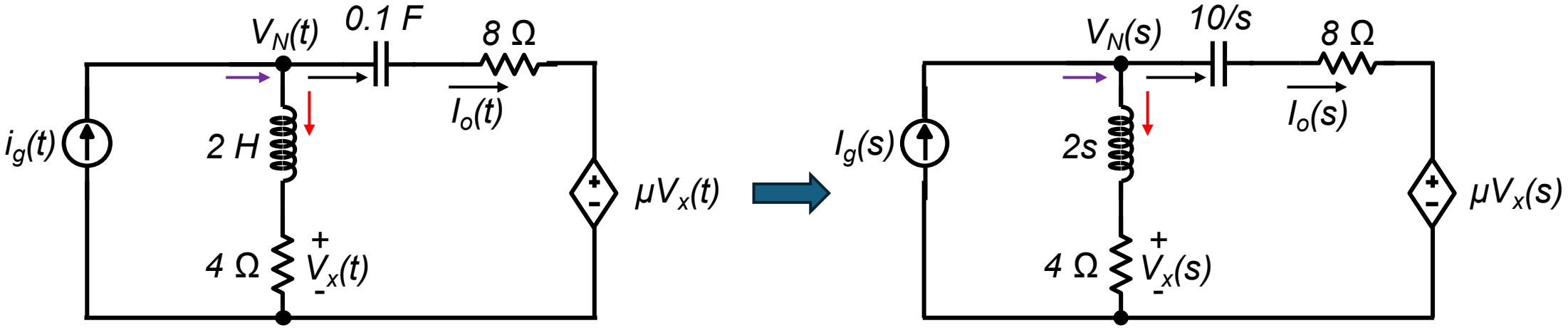


$$s_{1,2} = \frac{-(6-2\mu) \pm \sqrt{(6-2\mu)^2 - 20}}{2}$$

$$For\ stability, \frac{6-2\mu}{2} > 0\ , hence\ \ 0 < \mu < 3$$

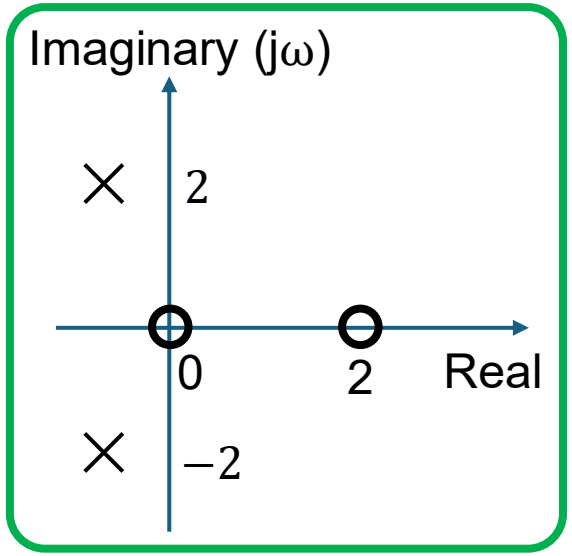
NOTE: $\boxed{L\{f(t) = e^{at}\} = \frac{1}{s-a}}$

*The exponential explodes if a > 0 & t > 0

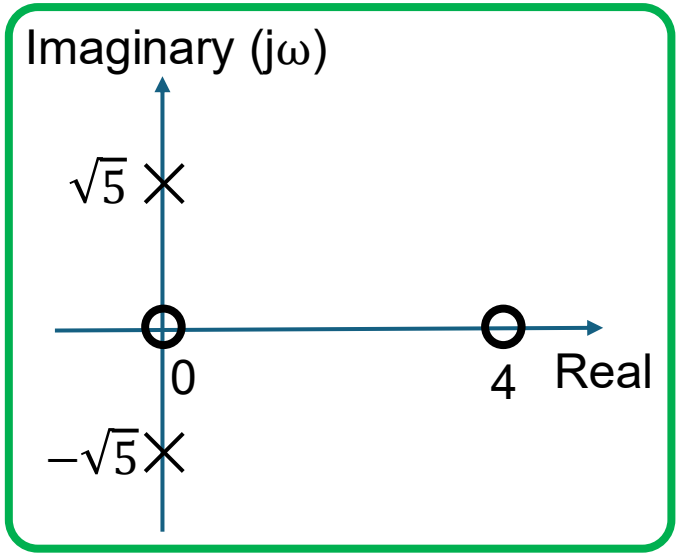
*At μ > 3, poles cross into the right half plane (RHP) and the system H(s) becomes unstable.

Concepts: Laplace Transform with Zero Initial Conditions + Stability

μ = 2

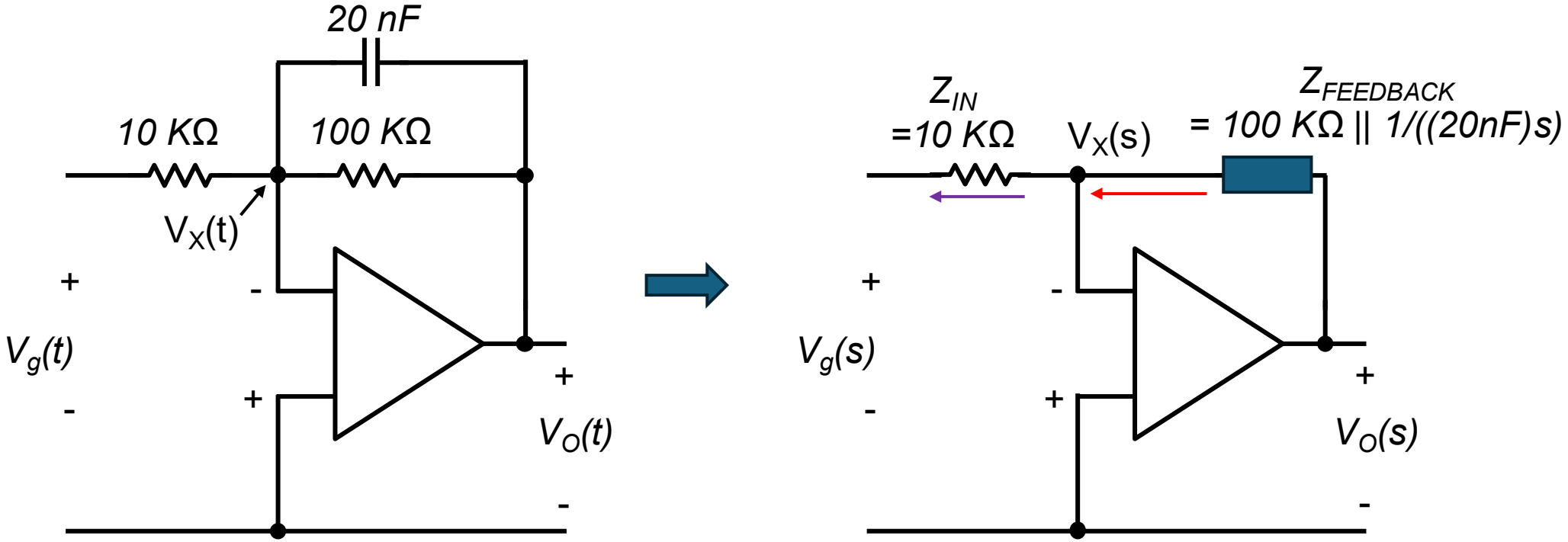


μ = 3



Concepts: Applying Laplace Transform to Circuit Analysis in Time Domain + Op Amp

Find the unit impulse response h(t) and the unit step response y(t), with zero initial conditions:



$$\frac{V_X(s) - V_g(s)}{Z_{IN} = 10{,}000} = \frac{V_O(s) - V_X(s)}{Z_{FEEDBACK} = 100{,}000 \| \frac{1}{s(20nF)}}$$

$$H(s) = \frac{V_O(s)}{V_g(s)} = -\frac{Z_{FEEDBACK}}{Z_{IN}} = -\frac{\frac{\frac{1}{Cs}(100{,}000)}{100{,}000 + \frac{1}{Cs}}}{10{,}000}$$

Concepts: Applying Laplace Transform to Circuit Analysis in Time Domain + Op Amp

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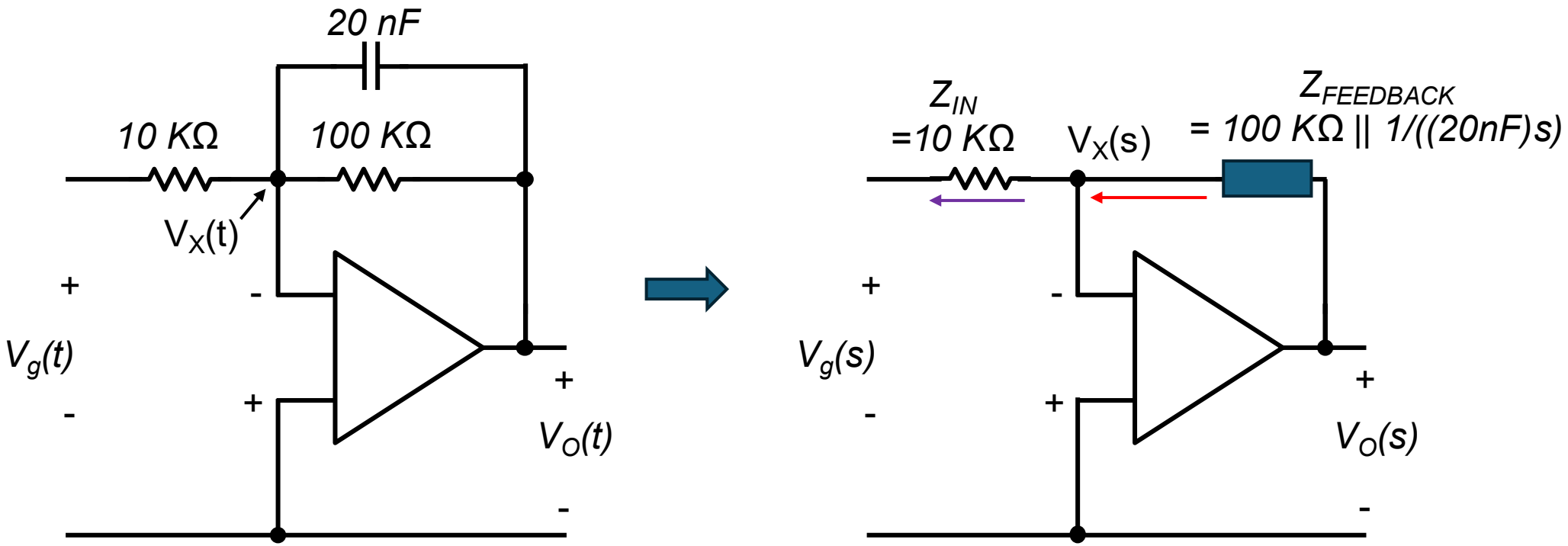


$$H(s) = -\frac{\frac{1}{Cs}(100{,}000)}{100{,}000 + \frac{1}{Cs}} = -\frac{10}{100{,}000Cs + 1} = -\frac{10}{100{,}000 \cdot 20 \times 10^{-9}s + 1} = -\frac{10}{2 \times 10^{-3}s + 1} = -\frac{5{,}000}{s + 500}$$

Concepts: Applying Laplace Transform to Circuit Analysis in Time Domain + Op Amp

Find the unit impulse response h(t) and the unit step response y(t), with zero initial conditions:

$$H(s) = -\frac{5{,}000}{s+500}$$

$\boxed{h(t) = -5{,}000\, e^{-500t}\, u(t)}$ This, by definition, is the impulse response.

For the unit step response y(t), find the Y(s) first then fine y(t) from Y(s) by Laplace Transforming Y(s) into y(t):

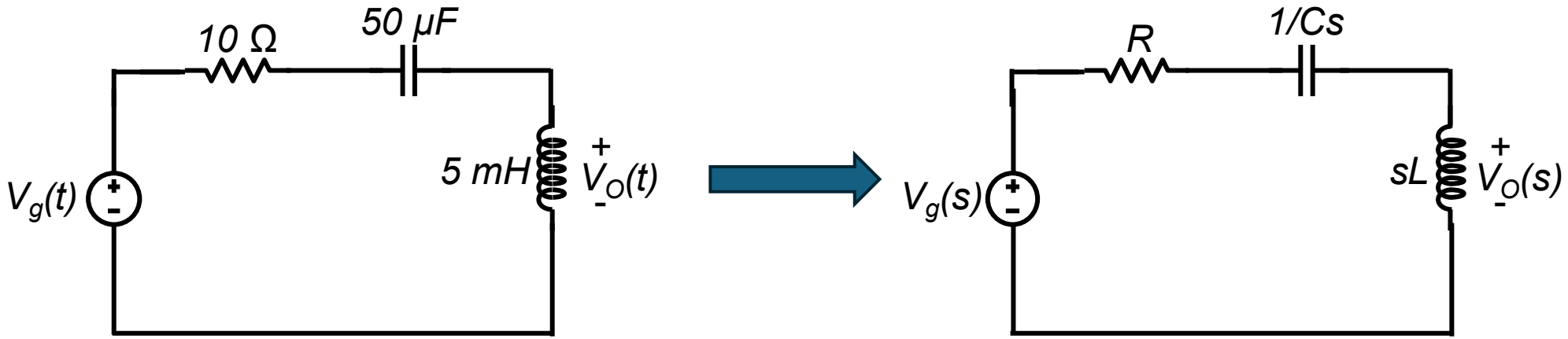
$$V_O(s) = H(s)V_g(s) = \left(-\frac{5000}{s+500}\right)\left(\frac{1}{s}\right) = -5000\left[\frac{A}{s} + \frac{B}{s+500}\right] = -5000\left[\frac{\frac{1}{500}}{s} + \frac{-\frac{1}{500}}{s+500}\right] = -10\left[\frac{1}{s} - \frac{1}{s+500}\right]$$

$$V_O(t) = \boxed{10[e^{-500t} - 1]\, u(t)} = y(t)$$

Concepts: Applying Laplace Transform to Circuit Analysis + "Steady-State"

$V_g(t)$ = 10 cos (2000 t) and the circuit is in the steady-state (zero initial conditions).

Find the steady-state output $V_{oss}(t)$:



$$V_O(s) = V_g(s)\left(\frac{sL}{R + \frac{1}{Cs} + sL}\right) = V_g(s)\left(\frac{LCs^2}{LCs^2 + RCs + 1}\right) = V_g(s)\left(\frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}}\right)$$

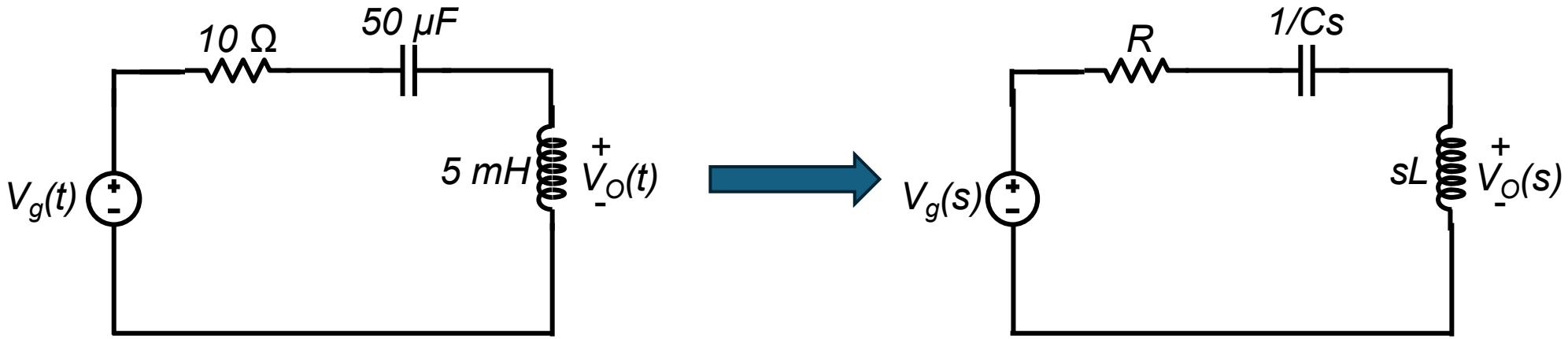
$$\frac{V_O(s)}{V_g(s)} = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{s^2}{s^2 + 2000\,s + 4 \cdot 10^6}$$

Recall:
$s = j\omega$, where $V_g(t)$ = 10 cos (2000 t) means that ω = 2000

Concepts: Applying Laplace Transform to Circuit Analysis + "Steady-State"

$V_g(t)$ = 10 cos (2000 t) and the circuit is in the steady-state (zero initial conditions).

Find the steady-state output $V_{oss}(t)$:



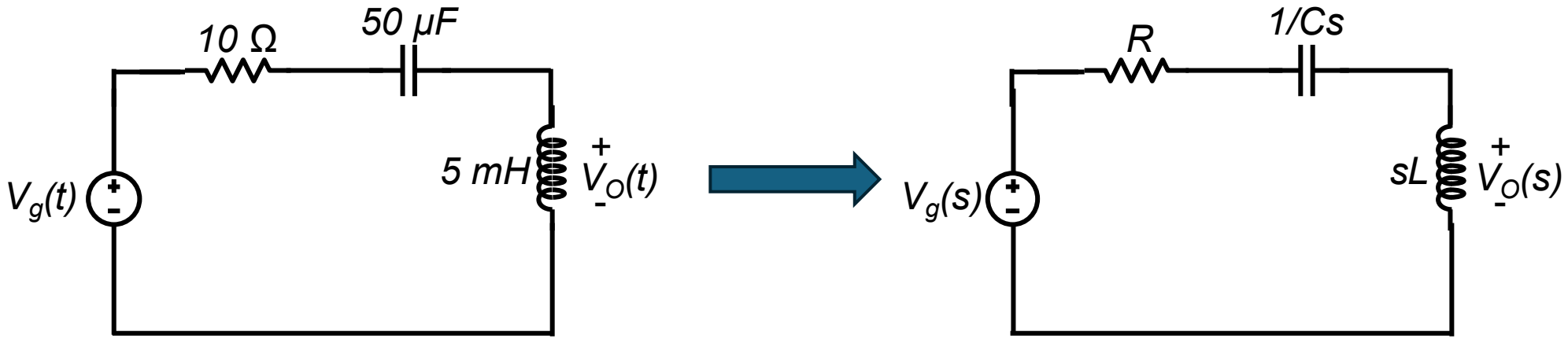
$$\frac{V_O(s)}{V_g(s)} = H(s) = \frac{s^2}{s^2 + 2000\,s + 4\cdot 10^6} \qquad H(s = j2000) = \frac{(j2000)^2}{(j2000)^2 + 2000(j2000) + 4\cdot 10^6} = j = 1\angle 90°$$

$$V_O(t) = |H(j2000)|\cos\big(\omega t + \angle H(j2000)\big) = 10\cos(\omega t + 90°) = \boxed{10\cos(2000t + 90°)}$$

Concepts: Applying Laplace Transform to Circuit Analysis + "Steady-State"

$V_g(t)$ = 10 cos (2000 t) and the circuit is in the steady-state (zero initial conditions).

Find the steady-state output $V_{oss}(t)$ **<u>if we set $V_g(t)$ = 5 V</u>**:



When the $V_g(t)$ = 5 V, $s = 0$ and hence *H(0)* = 0

$$\frac{V_O(s)}{V_g(s)} = H(s) = \frac{s^2}{s^2 + 2000\,s + 4 \cdot 10^6} \qquad H(s = 0) = 0$$

Or you can see that the capacitor will be open so the voltage at $V_O(t)$ will be 0.

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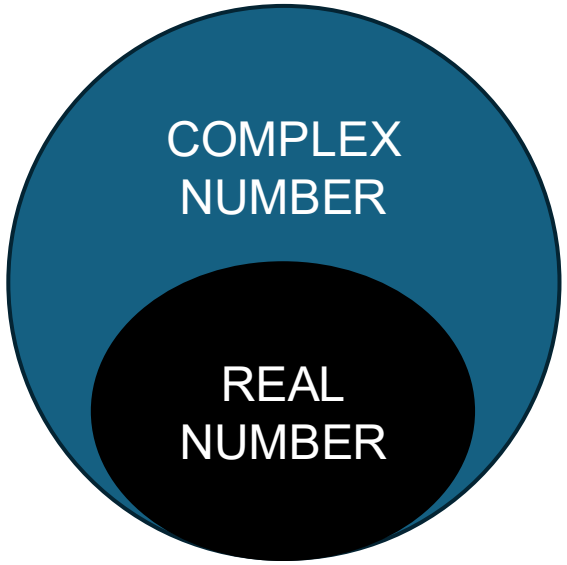
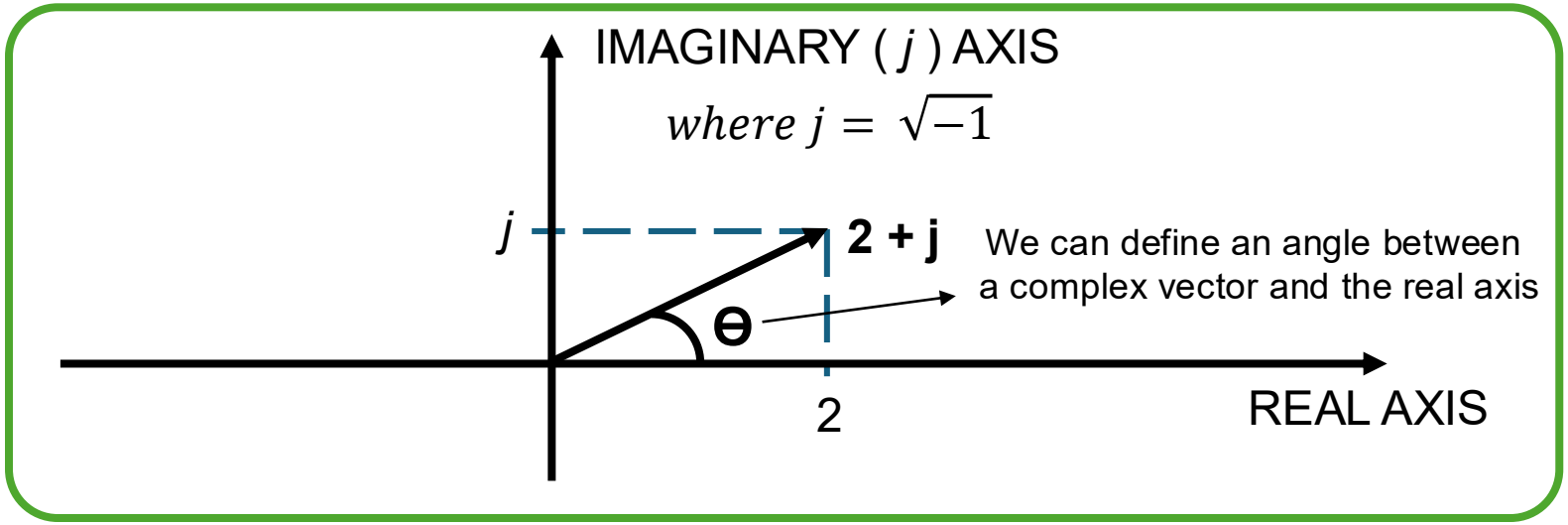
Zeros, poles, and stability

- Zeros, poles, and stability
- **Complex number and its relationship to phase angle**

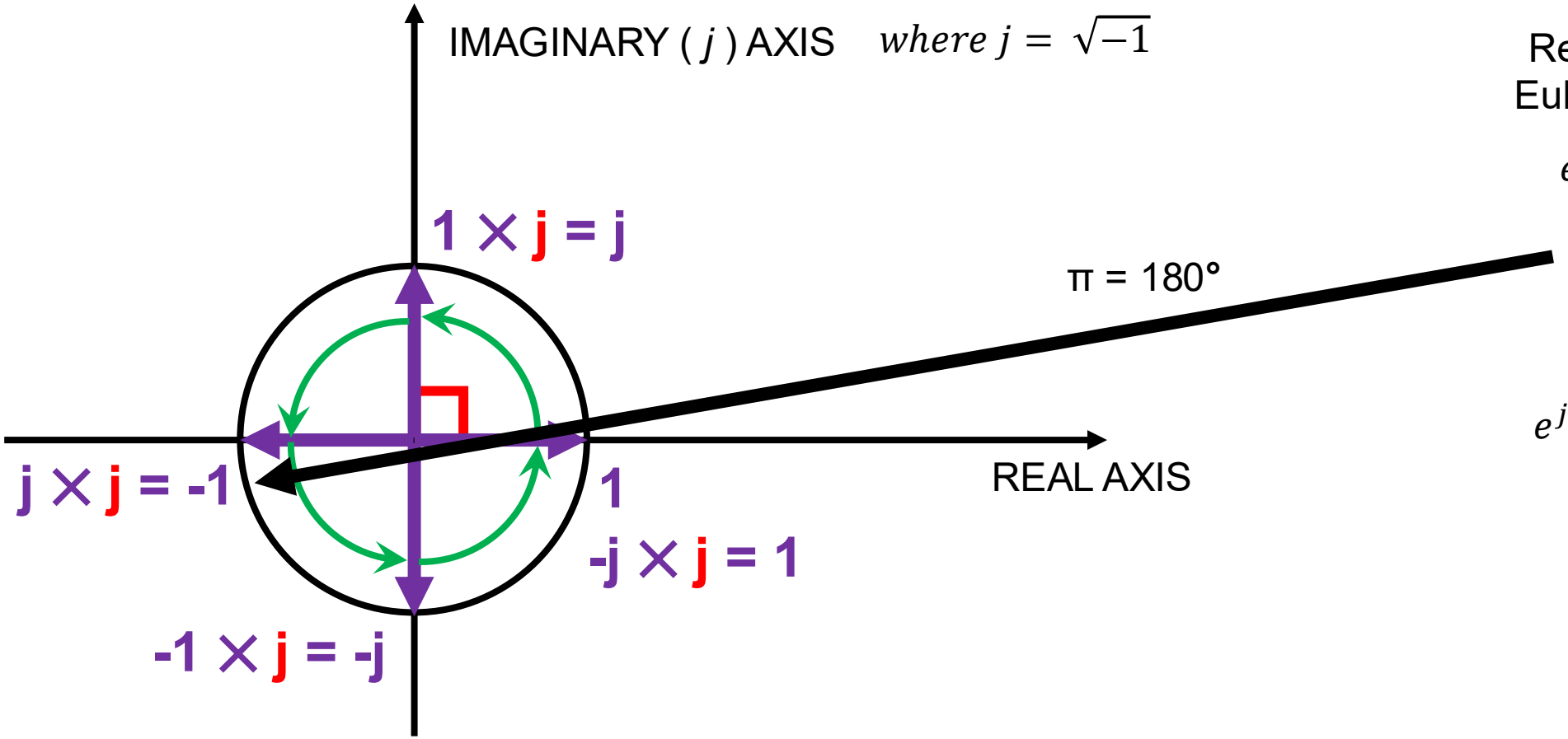
Complex number and its relationship to phase angle



Complex Plain



Complex number and its relationship to phase angle



× j = 90° phase shift

Relationship to Euler's Theorem

$$e^{j\pi} + 1 = 0$$

$$e^{j\pi} = -1$$

$$e^{j0} = 1 = e^{j2\pi}$$

$$e^{j\frac{\pi}{2}} = j$$

$$e^{j\frac{3\pi}{2}} = -j$$

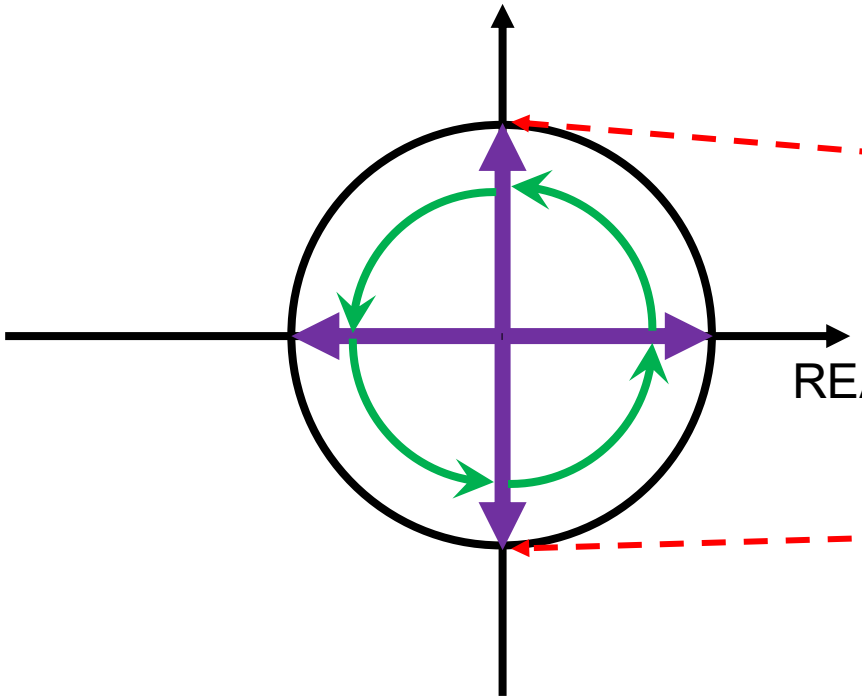
Complex number and its relationship to phase angle

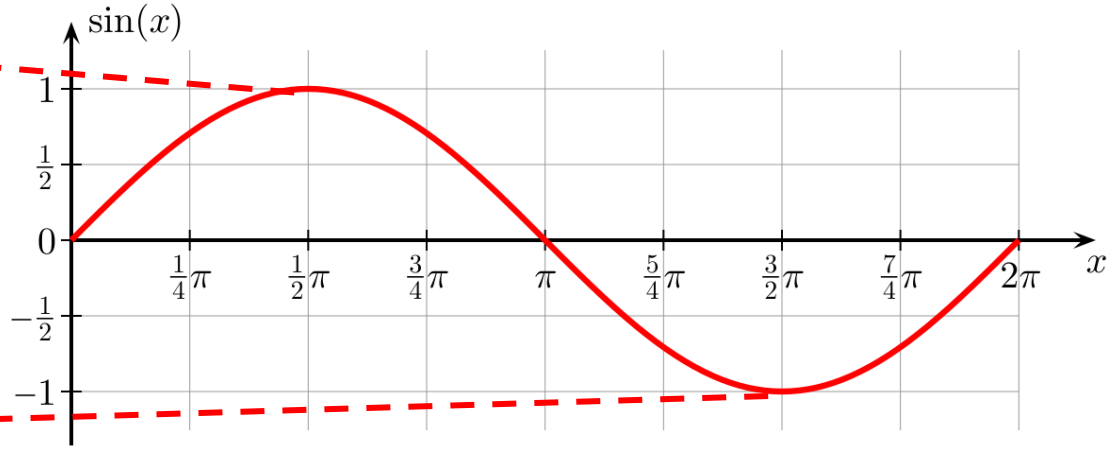
IMAGINARY (j) AXIS

$where\ j = \sqrt{-1}$

$$\sin(\omega t) = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

$$cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$





π = 180 °